MPFASS-Net: A Mamba Progressive Feature Aggregation Network With Self-Supervised for Remote Sensing Image Scene Classification

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***Abstract*—Focusing on the semantic gap problem of "large intraclass differences and small interclass differences" caused by the complex background and scale variations of interest targets in remote sensing image (RSI), a novel Mamba progressive feature aggregation network with self-supervised (MPFASS-Net) is designed, aiming to mine the multiscale global contextual information of RSI and capture its multigranularity semantic information, thereby further improving the accuracy of RSI scene classification. First, ResNet50 is adopted as the backbone network for extracting the multilayer raw features of RSI. Second, a cross-layer feature aggregation mechanism combining dual-cross attention (DCA) and Mamba is designed, and a hierarchical feature fusion module is constructed to obtain more discriminative multigranularity features, aiming to alleviate the effects of target scale variations in RSI. Third, two self-supervised functions based on class-center vectors are constructed to alleviate the semantic gap problem in RSI scene classification. Finally, by combining the ideas of deep supervision and classifier ensemble, a new decision-level data fusion module is further designed that can explore the complementarity of hierarchical fusion features, thereby obtaining an ensemble classifier with stronger generalization ability. Comparative experimental results based on three datasets, i.e., AID, NWPU, and UCM, show that the classification accuracies of the designed MPFASS-Net model are 97.62%, 94.93%, and 99.87%, respectively, and are superior to other state-of-the-art methods.**

***Index Terms*—Decision level classifier ensemble, hierarchical feature aggregation, remote sensing image (RSI) scene classification, self-supervised learning (SSL).**

I. Introduction

IN RECENT years, with the rapid development of remote sensing and communication technology, various high-precision sensors have been manufactured and installed on satellites or high-altitude aircraft, and the collected remote sensing images (RSI) are widely used in urban planning, intelligent transportation, environmental monitoring, and other fields [1]. However, due to the increasing number of RSI, it is of great significance to use computers to automatically classify RSI scenes to improve the efficiency of observing the Earth's surface based on RSI [2].



Fig. 1. Sample images from NWPU. (a) Large intraclass differences, i.e., these four images belong to the same category, but their visual features are very dissimilar. (b) Small interclass differences, i.e., these 4 images belong to different categories, but their visual features are very similar.

RSI scene classification, as a popular research branch in the field of computer vision, the early methods are mainly been based on hand-crafted features. Their common idea is to extract the underlying visual features, such as color [3], texture, and shape [4] of RSI, and then combine them with a traditional machine learning method (e.g., SVM) to achieve RSI scene classification. However, as shown in Fig. 1, due to the semantic gap problem of "large intraclass differences and small interclass differences" in visual properties of RSI, hand-crafted features are difficult to effectively represent the complex and diverse semantic information in RSI, leading to low accuracy in RSI scene classification [2]. Due to the outstanding performance of convolutional neural networks (CNN) [5] in image classification tasks, the current mainstream methods for RSI scene classification are based on CNN. For example, Hu et al. [6] propose a variational self-distillation network, in which the class entanglement information from the prediction vector acts as a supplement to the category information. Shi et al. [7] constructed efficient pooled channel attention and efficient convolutional coordinate attention modules in response to the fact that the complex backgrounds and small objects in RSI pose a challenge to CNN, thus designing

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a novel multibranch fusion attention network (MBFANet) for RSI scene classification.

Although CNN's convolution operations are beneficial for capturing local detail features and have shown excellent performance in RSI scene classification, they still have the disadvantage of insufficient modeling ability for RSI global contextual information. Whereas the visual transformer (ViT) [8], due to its ability to capture global contextual information in an image, then it is also used for RSI scene classification tasks. For example, Ma et al. [9] designed a homo-heterogeneous transformer learning (HHTL) framework, which aims to explore contextual information in RSI and compensate for the shortcomings of CNN's global modeling capabilities. Lv et al. [10] designed a spatial channel feature preserving ViT (SCViT) model, i.e., by introducing a lightweight channel attention module, which takes into account both the local detail information of the RSI as well as the global contextual information. In summary, although various RSI scene classification methods have their own advantages, when facing the semantic gap problem of "large intraclass differences and small interclass differences" caused by the complex background of RSIs and scale variations of the key target, the main challenges to further improve the accuracy of RSI classification are follows.

1) Multilevel local detail features in the backbone network are yet to be effectively aggregated to enhance the multigranularity semantic discriminative capability of the fused features.

2) Transformer suffers from the disadvantages of large parameters and high-computational cost in modeling the global context information of RSI.

3) These methods do not take specific measures to improve the robustness of feature extraction and the generalization ability of the model.

Therefore, focusing on the above challenges present in the RSI scene classification task, a novel Mamba progressive feature aggregation network with self-supervised (MPFASS-Net) is designed in this article, i.e., which is an end-to-end feature-level and decision-level dual data fusion network, aimed at mining the hierarchical interdependence among multiscale feature maps of RSI and utilizing the complementarity among multiclassifiers to further improve the accuracy of RSI scene classification. As shown in Fig. 2, the MPFASS-Net model mainly consists of four modules, i.e., multiscale feature extraction (MSFE), progressive multiscale feature aggregation (PMSFA), orthogonal clustering self-supervised learning (OCSSL), and decision-level classifiers ensemble (DLCE) modules. Finally, the main contributions of this article are summarized as follows.

1) In the PMSFA module, a cross-layer feature aggregation mechanism combining dual-cross attention (DCA) and Mamba is designed, i.e., using Mamba's excellent global context modeling capability to perform hierarchical cross-layer iterative fusion on multiscale feature maps in the backbone network, aiming to obtain more discriminative multigranularity semantic features and mitigate the effects of key target scale variations in RSI.

2) In the OCSSL module, two self-supervised learning (SSL) functions based on some learnable class-center vectors are designed to constrain the training of the model, aiming to refine the multiscale fused features and alleviate the semantic gap problem of "large intraclass differences and small interclass differences" in RSI scene classification.

3) In the DLCE module, the idea of deep supervision and classifier ensembles is combined to design a decision-level adaptive data fusion strategy, i.e., integrating classifiers from different levels of fused features to enhance the generalization ability of the designed model by exploiting their complementarities.

II. Related Work

A. RSI Scene Classification

As an important research direction in the field of computer vision, today's RSI scene classification methods are mainly based on deep learning, and for the multiscale problem of interest targets in RSI, many methods based on multilayer feature fusion and attention mechanisms have been proposed. For example, Yang et al. [11] designed a semantic-aware graph network (SAGN) for RSI classification, which utilizes the unstructured relationships between different semantics to mine multiscale features of RSI. Zhang and Wang [12] designed a progressive feature fusion model based on graph convolution for RSI scene classification. The highlight of this framework is the combination of GCN and CNN, which can more effectively capture the potential contextual relationships of RSI and improve classification accuracy. Wang et al. [13] designed an adaptive channel dimensionality reduction multilevel feature fusion network for remote sensing scene classification. Hou et al. [14] designed a contextual spatial channel attention network for RSI scene classification. Sitaula et al. [15] designed a new plug-and-play multiscale attention feature extraction model based on two-level multiscale convolution with skip connections, which generates discriminative/salient information at a deeper/finer level of RSI. In summary, although CNN methods driven by GCN and attention have improved the accuracy of RSI scene classification, they still have the disadvantage of insufficient perception of multigranularity semantic information and hierarchical features in RSI.

In order to better explore the multigranularity semantic information of RSI and further improve its classification accuracy, more semantic-aware fusion models have been designed, for example Zhao et al. [16] designed a collaborative dictionary learning network (CDLNet) for RSI scene classification by analyzing the specific contribution of local features to global semantics. Song et al. [17] designed a semantic-aware fusion network (SAF-Net), which mainly contains semantic-aware and multilayer feature fusion modules to better extract the multigranularity features and improve the RSI scene classification accuracy. Zhao et al. [18] designed a new efficient multisample contrastive network (EMSCNet), which combines metric learning with knowledge distillation to improve RSI classification accuracy by integrating knowledge from multiple samples. Yang et al. [19] designed an RSI classification method based on large visual models (LVMs), which efficiently explores the multiscale semantic features of RSI by attaching learnable prompt tokens to the input tokens to improve the accuracy of RSI classification.

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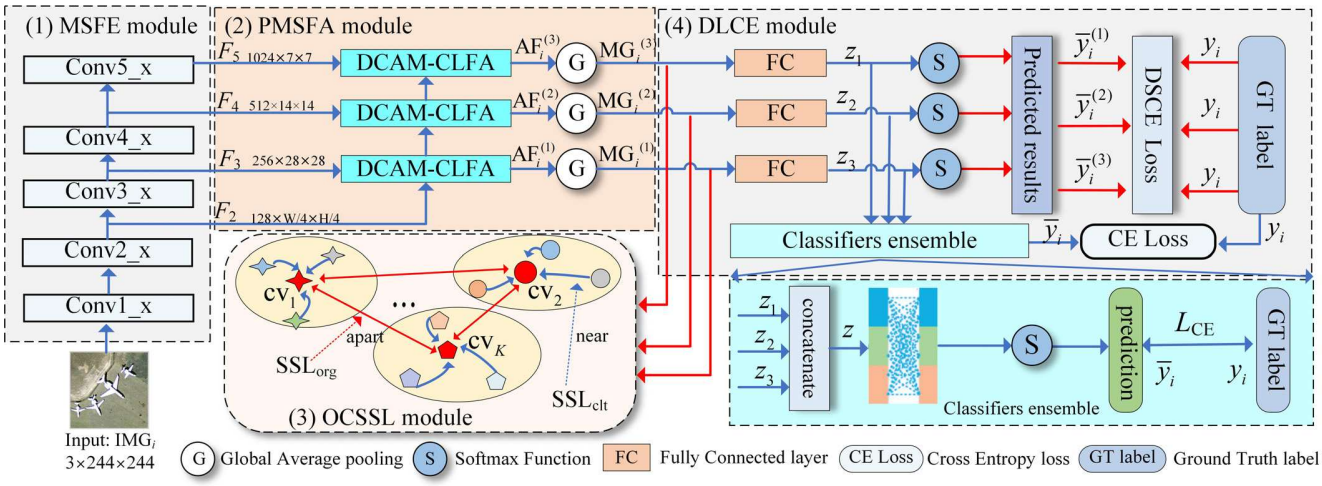


Fig. 2. Overview of the designed MPFASS-Net. The input image is first processed by the MSFE module to obtain 4-layer multiscale features, then cross-layer aggregation is performed in the PMSFA module using the designed DCAM-CLFA mechanism, then the 3-layer aggregated features are refined by the OCSSL module, and finally sent to the DLCE module for classifier ensemble to enhance the generalization capability.

B. State Space Model

State space models (SSMs) originated from the classical Kalman filter model, which has recently been very successfully applied in natural language processing and computer vision due to its expertise in capturing long-range dependencies of sequential information, and with linear computational complexity [20]. For example, Gu et al. [21] designed a new parameterization method based on low-rank structure correction, and constructed a new structured SSM called S4. By designing a multiple-input multiple-output SSM and a parallel scanning strategy, Smith et al. [22] constructed a new SSM model called S5 based on the S4 model. Meanwhile, the S6 model [23] introduces a data-dependent SSM layer to S4, which integrates a selective method using parallel scanning. It adopts convolution calculation and near-linear calculation methods, which demonstrate excellent performance in long sequence processing. Subsequently, Mamba [24], as a novel SSM model for natural language processing applications, incorporated time-varying parameters into SSM and proposed a hardware-aware algorithm. Since Mamba is not only very efficient in training and inference, but also has excellent scalability, it is expected to be an alternative to Transformer in language modeling.

Recently, many researchers have introduced Mamba into the field of computer vision. Since Mamba was originally applied only to continuous 1-D sequences and could not directly process visual data with 2-D structure such as images, then inspired by ViT [8], some visual Mamba models (e.g., VMamba [20] and ViM [25]) have be designed, i.e., to reduce the discrepancy between 1-D sequential scanning and 2-D planar traversal, VMamba designs a 2-D selective scanning (SS2D) module, which traverses the image spatial by four-way scanning (i.e., top left, bottom right, top right and bottom left), thus solving the spatial position sensitivity problem to be faced by 2-D planar traversal, while ViM uses forward and backward bi-directional scanning to enable the model to perceive each part of the image, thus obtaining global information about the image. Finally, based on the VMamba/ViM model, Mamba has been widely used in many visual fields, such as Liu et al. [26] designed a ViM-based RSCaMa model for change description of RSIs. Chen et al. [27] proposed RSMamba for RSI classification by combining shuffle with forward and backward scanning. Ma et al. [28] proposed a novel two-branch network, RS3Mamba, for semantic segmentation of RSI.

C. Self-Supervised Learning

SSL is a popular research direction in the field of machine learning in recent years. It aims to improve the feature extraction ability of models for unlabeled data by designing proxy tasks to mine the representation characteristics of the data itself as supervised information [29]. For example, Bountos et al. [30] proposed using self-supervised contrastive learning (CL) to learn high-quality visual representations hidden in unlabeled ImSAR data. Xue et al. [31] proposed a generative SSL framework for land cover classification of multimodal RSI. Yang et al. [32] proposed a self-supervised method for hyperspectral image classification using a pre-trained model fine-tuned with a large amount of unlabeled data. Bi et al. [33] proposed a ViT model combining supervised CL, which extends the self-supervised contrastive method to fully supervised situations. It can mine the label information of RSI in the embedding space and improve robustness against common image damage. Li et al. [34] proposed a multivariate ensemble SSL (MES2L) framework for few-shot RSI classification to address the difficulties in obtaining and labeling remote sensing data.

In addition, the latest SSL methods include Zhang et al. [35] proposed a relationship-aware contrastive SSL (ReCo) method, Tao et al. [36] proposed a generic self-supervised framework SLMR, and Manna et al. [37] proposed a self-supervised representation learning (SKID) framework, which have achieved better performance in tasks such as image classification, multimedia recommendation, and knee joint injury diagnosis.

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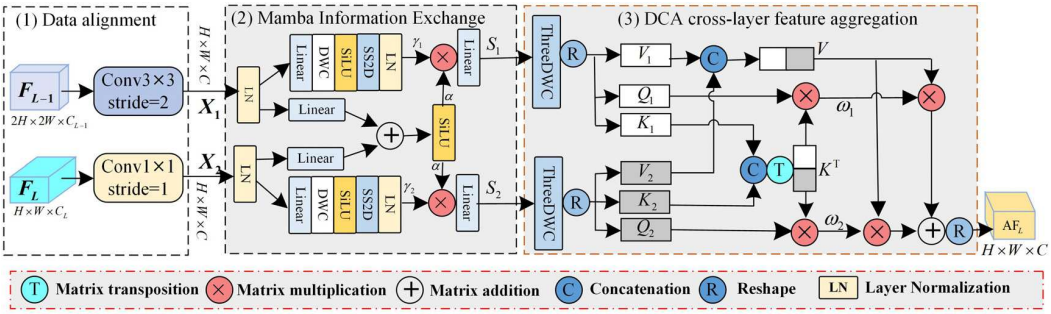


Fig. 3. Schematic of the designed DCAM-CLFA mechanism.

III. Designed MPFASS-Net Model

Focusing on the semantic gap problem of "large intraclass differences and small interclass differences" caused by the complex background of RSI and the scale variation of key targets, as shown in Fig. 2, a novel MPFASS-Net model is designed, which mainly includes four modules, i.e., MSFE, PMSFA, OCSSL, and DLCE. The aim is to utilize the global contextual modeling capability of multilayer Mamba to effectively capture multigranularity semantic features of RSI, and, combined with the designed OCSSL and CLCE modules, to alleviate the semantic gap problem and enhance the generalization ability of the designed model.

A. MSFE Module Design

Since ResNet50 [5] has excellent feature extraction capability in computer vision tasks such as object detection and image classification, its five residual blocks with an average pooling layer are chosen to construct an MSFE backbone network for the designed MPFASS-Net. Let IMG_i denotes any RSI input to the network, and the feature maps outputted by the 2nd to 5th residual blocks (i.e., Conv2_x to Conv5_x) are denoted as F_2, F_3, F_4, and F_5, which are denoted as

$$F_L = \mathrm{Conv}L_x\left(\mathrm{IMG}_i\right), \quad L = 2,3,4,5 \tag{1}$$

where $F_L \in \mathbb{R}^{(H/2^L)\times(W/2^L)\times C_L}$ (L = 2,3,4,5) represents extracted feature maps with different scales, H and W represent the height and width of the input image, respectively.

B. PMSFA Module Design

To effectively capture the multigranularity information of RSI and alleviate the interference of scale variation of interest targets, a PMSFA module is designed to progressively fuse the last 4 layers' feature maps obtained from the backbone network. Specifically, as shown in (1), let $F = \{F_L\}_{L=2}^{5}$ denotes the last 4-layer feature maps with different scales; then, as shown in the PMSFA module in Fig. 2, the cross-layer iterative feature fusion process from bottom-to-top can be formalized as

$$\begin{cases} \mathrm{AF}_i^{(1)} = \mathrm{DCAM\text{-}CLFA}\left(F_2, F_3\right) \\ \mathrm{AF}_i^{(2)} = \mathrm{DCAM\text{-}CLFA}\left(F_4, \mathrm{AF}_i^{(1)}\right) \\ \mathrm{AF}_i^{(3)} = \mathrm{DCAM\text{-}CLFA}\left(F_5, \mathrm{AF}_i^{(2)}\right) \end{cases} \tag{2}$$

where, DCAM-CLFA(·) denotes the designed Dual-Cross Attention Mamba Cross-Layer Feature Aggregation (DCAM-CLFA) mechanism. $\mathrm{AF}_i^{(m)}, m = 1,2,3$ denotes the cross-layer fusion result.

Let $F_L \in \mathbb{R}^{H\times W\times C_L}$ and $F_{L-1} \in \mathbb{R}^{2H\times 2W\times C_{L-1}}$ represent any two layers of feature maps to be aggregated. In order to interactively fuse them and enhance the multigranularity semantic representation capability, as shown in Fig. 3, DCAM-CLFA mainly consists of three steps, i.e., data alignment (DA), mamba information exchange (MIE), and DCA cross-layer feature aggregation.

1) Data Alignment: To ensure that the two feature maps to be fused have the same spatial resolution and number of channels, a convolution operation is used to align F_{L-1} and F_L, i.e., using a 3×3 convolutional layer $f_{3\times 3}(\cdot)$ with a stride of 2 to process F_{L-1}, while using another 1×1 convolutional layer $f_{1\times 1}(\cdot)$ with a stride of 1 to process F_L, which is formulated as

$$\begin{cases} X_1 = f_{3\times 3}\left(F_{L-1}\right) \\ X_2 = f_{1\times 1}\left(F_L\right) \end{cases} \tag{3}$$

where X_1 and $X_2 \in \mathbb{R}^{H\times W\times C}$ denote the results of the DA process; H, W, and C (i.e., C = 256) denote height, width, and number of channels, respectively.

2) Mamba Information Exchange: To utilize Mamba's long-range modeling capability to capture global contextual information of X_1 and X_2, following the VMamba method [20], layer normalization (LN) is first applied to process X_1 and X_2 separately, and then the results of LN are fed into the three branches of VSSM for information exchange. Specifically, as shown in Fig. 3, in the middle branch of VSSM, two linear layers are first used to expand the channel numbers of X_1 and X_2 to $2C$, then element-wise addition and SiLU activation are performed to obtain the shared gating factor α, i.e.,

$$\alpha = \mathrm{SiLU}\left(\mathrm{Linear}\left(\mathrm{LN}\left(X_1\right)\right) + \mathrm{Linear}\left(\mathrm{LN}\left(X_2\right)\right)\right). \tag{4}$$

Meanwhile, the top and bottom branches also first use the linear layer for channel extension, and then sequentially perform depth-wise convolution (DWC), SiLU activation, SS2D layer [20], and LN. The results obtained are denoted as

$$\begin{cases} \gamma_1 = \mathrm{LN}\left(\mathrm{SS2}D\left(\mathrm{SiLU}\left(\mathrm{DWC}\left(\mathrm{Linear}\left(\mathrm{LN}\left(\mathrm{X}_1\right)\right)\right)\right)\right)\right) \\ \gamma_2 = \mathrm{LN}\left(\mathrm{SS2D}\left(\mathrm{SiLU}\left(\mathrm{DWC}\left(\mathrm{Linear}\left(\mathrm{LN}\left(\mathrm{X}_2\right)\right)\right)\right)\right)\right) \end{cases} \tag{5}$$

where, DWC(·) and LN(·) denote 3×3 DWC and LN, respectively. Notably, SS2D(·) denotes the core component of VSSM, i.e., the SS2D module [25], as shown in Fig. 4.

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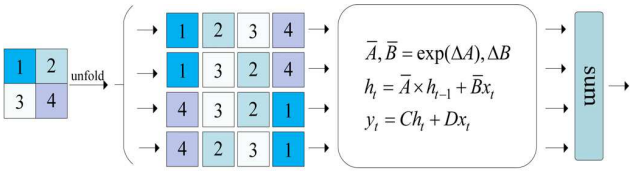


Fig. 4. Schematic of the SS2D module.

Finally, the obtained results γ_1 and γ_2 are the Hadamard product with the gating factor α, respectively, for information aggregation and exchange. Next, the channel is adjusted by a linear layer to be consistent with the input, and then, the obtained two sequences are denoted as

$$\begin{cases} S_1 = \text{Linear}(\gamma_1 \odot \alpha) \\ S_2 = \text{Linear}(\gamma_2 \odot \alpha) \end{cases} \tag{6}$$

where $\odot$ denotes Hadamard product. After MIE, S_1 and $S_2 \in \mathbb{R}^{H \times W \times C}$ represent the results corresponding to X_1 and X_2, respectively.

3) DCA Cross-Layer Feature Aggregation: To fuse S_1 with S_2 and enhance its semantic expression ability, the designed DCA cross-layer feature aggregation mechanism is:

First, two groups of DWCs are utilized to process S_1 and S_2, respectively, thereby (Q_1, K_1, V_1) and (Q_2, K_2, V_2) can be obtained after the reshape, i.e.,

$$\begin{cases} Q_1, K_1, V_1 = \text{Reshape}(\text{ThreeDWC}(S_1)) \\ Q_2, K_2, V_2 = \text{Reshape}(\text{ThreeDWC}(S_2)) \end{cases} \tag{7}$$

where ThreeDWC$(\cdot)$ and Reshape$(\cdot)$ denote three 3×3 DWCs and reshaping, and $Q_1, K_1, V_1, Q_2, K_2, V_2 \in \mathrm{R}^{HW \times C}$ represent the query, key, and value vectors after reshape processing.

Next, to implement information interaction between (Q_1, K_1, V_1) and (Q_2, K_2, V_2), the following concatenation operation is performed, i.e.,

$$V = \text{Concat}(V_1, V_2), \quad K = \text{Concat}(K_1, K_2) \tag{8}$$

where $K, V \in \mathrm{R}^{2HW \times C}$ denotes the concatenate results. Then, Q_1 and Q_2 are multiplied by the transpose of K, respectively, and after being processed by the Softmax function, we can obtain the DCA weights $\omega_1, \omega_2 \in \mathrm{R}^{HW \times 2HW}$, i.e.,

$$\begin{cases} \omega_1 = \text{Softmax}\left(Q_1 K^{\mathrm{T}} / \sqrt{C}\right) \\ \omega_2 = \text{Softmax}\left(Q_2 K^{\mathrm{T}} / \sqrt{C}\right). \end{cases} \tag{9}$$

Finally, cross-layer feature aggregation guided by DCA can be formalized as

$$\text{AF}_L = \text{Reshape}(\omega_1 V + \omega_2 V) \tag{10}$$

where $\text{AF}_L \in \mathrm{R}^{H \times W \times C}$ denotes the Mamba cross-layer feature fusion result corresponding to the Lth layer. Notably, in (9), taking ω_1 as an example. Compared to ordinary cross-attention, it has the special property of "mixed attention" [38]. This is because K is concatenated by K_1 and K_2, i.e., the dot product of Q_2 and the first half of K is obtained as "self-attention," while the dot product of Q_2 and the last half of K is obtained as "cross-attention."

C. OCSSL Module Design

Let IMG_i denotes any a training image, which after being processed by MSFE and PMSFA modules, as shown in (2), three different fused features $\text{AF}_i^{(L)} \in \mathbb{R}^{C \times H \times W} (L = 1, 2, 3)$ can be obtained, then $\text{AF}_i^{(L)}$ after global average pooling (GAP), corresponding *multigranularity features* $\text{MG}_i^{(L)} \in \mathbb{R}^{1 \times C}$ are obtained, i.e.,

$$\text{MG}_i^{(L)} = \text{GAP}\left(\text{AF}_i^{(L)}\right), \quad L = 1, 2, 3. \tag{11}$$

Although $\text{MG}_i^{(L)}$ is derived from Mamba multiscale feature fusion, which enhances its ability to represent the multigranularity semantics of RSI, it still does not specifically consider the semantic gap problem in RSI scene classification. Therefore, an OCSSL module is designed based on the human mindset in image classification tasks [39], i.e., assuming that there are some potential "class center vectors" in the semantic space, which correspond to the semantic information of the target-of-interest identification for each class of RSI. Specifically, the designed OCSSL aims to utilize the multigranularity feature $\text{MG}_i^{(L)}$ of the training images to automatically learn a "class-center vector" $\text{cv}_k \in \mathbb{R}^{1 \times C}$ for each class of RSI, and arranging all class center vectors in rows is called *class-center matrix* $\text{CV} = [\text{cv}_k]_{N_c \times C}$, where N_c denotes the number of RSI categories. Obviously, cv_k in the CV is a C-dimensional row vector, which corresponds to the *class center vector* of the kth category RSI.

Let $D = \{(\text{IMG}_i, y_i)\}_{i=1}^{\text{BS}}$ represent any batch of training images, where BS represents batch-size, IMG_i and y_i represent the ith image and its one-hot ground-truth (GT) label, respectively. To obtain a set of excellent class center vectors to enhance the discriminative ability of multigranularity features, two SSL loss functions have been constructed based on the class center matrix CV, i.e.,

1) Orthogonal SSL Function: Due to every row vector in the class center matrix, CV represents a class-center vector corresponding to a category of RSI, which is used to describe the multigranularity semantic information of various types of RSIs, respectively. If all the class center vectors are orthogonal to each other, it means that they are distributed in different regions of the semantic space, which is more conducive to the classification of the RSI scenes. Therefore, the dot product between any two class center vectors in the CV is computed to obtain the mutual correlation matrix R, and then the orthogonal SSL loss function based on the class center vectors is defined as

$$\begin{cases} \text{SSL}_{\text{org}}(D) = \|R\|_2 \\ R = \left[r_{i,j:} = < \text{cv}_i, \text{cv}_j >\right]_{i=1, j=1, i \neq j}^{N_c \times N_c} \end{cases} \tag{12}$$

where $r_{i,j}$ denotes the result of the dot product between the ith class center vector cv_i and the jth class cv_j, $\|\cdot\|_2$ denotes the $L2$ norm. Obviously, since i cannot be equal to j, the values on the diagonal of R are all 0, thereby SSL_{org} will drive the values on the off-diagonal of the matrix R to converge to 0, i.e., making the individual class center vectors orthogonal to each other and more diverse. Notably, the dot product is essentially determined by the angle between two vectors, i.e., if two vectors are perpendicular (orthogonal) to each other,

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then the dot product is 0, indicating that they are completely unrelated.

2) Clustered SSL Functions: In the semantic space, to facilitate RSI scene classification, it is desirable for multigranularity features corresponding to RSIs of the same category to be tightly clustered together, while far away from other categories of RSIs, and the constructed clustered SSL loss function is defined as

$$\mathrm{SSL}_{\mathrm{clt}}(D) = \sum_{i=1}^{\mathrm{BS}} \frac{\|\mathrm{MG}_i - \mathrm{cv}_{y_i}\|_2}{\sum_{k=1,k\neq y_i}^{N_c} \|\mathrm{MG}_i - \mathrm{cv}_k\|_2} \tag{13}$$

where $\|\cdot\|_2$ denotes the computation of the Euclidean distance between MG_i and the class center vector cv. cv_{y_i} denotes the class center vector corresponding to the RSI image of class y_i. Obviously, in the process of model training, to minimize $\mathrm{SSL}_{\mathrm{clt}}$, we need to drive the molecular term to minimize, i.e., the Euclidean distance between MG_i and the corresponding class center vector cv_{y_i} tends toward 0. On the contrary, it is also necessary to maximize the denominator term, i.e., MG_i should be far away from all class center vectors corresponding to other categories [2].

D. DLCE Module Design

Let $D = \{(\mathrm{IMG}_i, y_i)\}_{i=1}^{\mathrm{BS}}$ represents any batch of training images, where BS denotes batch size, IMG_i and y_i represent the ith image and its label, respectively. As shown in (2), after iterative fusion through the DCAM-CLFA mechanism in the PMSFA module, each image can obtain three different "aggregation features," i.e., $\mathrm{AF}_i^{(1)}, \mathrm{AF}_i^{(2)}$, and $\mathrm{AF}_i^{(3)}$. Usually, the shallow layers contain the texture structure information of the image, while the high layers represent the semantic information of the image. Therefore, inspired by deep supervision [40] and classifier ensemble [41], this section designs a novel DLCE module, aiming to fully utilize the complementarity of the three-layer fusion features, thereby avoiding gradient vanishing and enhancing the model's generalization ability. Specifically, the DLCE module is mainly constructed by two components, i.e., deep supervision and classification result ensemble. Their detailed principles are as follows:

1) Deep Supervision: For each aggregation feature $\mathrm{AF}_i^{(m)}$ from the training sample (IMG_i, y_i) in D, which is processed by the GAP shown in (11) to obtain the multigranularity feature $\mathrm{MG}_i^{(m)}$, and then fed into a deep supervised classifier $f_{\mathrm{DS}}^{(m)}(\cdot)$, and the predicted label is denoted as

$$\begin{cases} \bar{y}_i^{(m)} = f_{\mathrm{DS}}^{(m)}\left(\mathrm{AF}_i^{(m)}\right) = \sigma\left(z_i^{(m)}\right), \quad m = 1,2,3 \\ z_i^{(m)} = \mathrm{FC}\left(\mathrm{MG}_i^{(m)}\right) = \mathit{ReLU}\left(\mathrm{MG}_i^{(m)} \times W_m + b_m\right) \end{cases} \tag{14}$$

where $\bar{y}_i^{(m)} \in \mathrm{R}^{N_c}$ denotes the predicted probability vectors over N_c semantic categories, $\mathrm{FC}(\cdot)$, $\mathrm{ReLU}(\cdot)$, and $\sigma(\cdot)$ denote the fully connected (FC) layer, ReLU activation, and Softmax function, respectively, $W_m \in \mathrm{R}^{C\times N_c}$ and $b_m \in \mathrm{R}^{1\times N_c}$ denote the learnable transform matrix and bias vector for the FC layer.

Meanwhile, to conduct model training, based on the two SSL functions described in (12) and (13), and combined with the deep supervised cross entropy (DSCE) loss, the total loss function L_{Total} is defined as follows:

$$\begin{cases} L_{\mathrm{Total}}(D) = L_{\mathrm{DSCE}}(D) + \mathrm{SSL}_{\mathrm{org}}(D) + \mathrm{SSL}_{\mathrm{clt}}(D) \\ L_{\mathrm{DSCE}}(D) = \sum_{i=1}^{\mathrm{BS}} \sum_{m=1}^{3} -y_i \log\left(\bar{y}_i^{(m)}\right) \end{cases} \tag{15}$$

where y_i and $\bar{y}_i^{(m)}$ denote the GT label and the predicted label of the ith image, respectively.

2) Classification Results Ensemble: As shown in the "classifiers ensemble" submodule in the bottom-right corner of Fig. 2, it takes the outputs z_1, z_2, and z_3 of three "deep supervised classifiers" as inputs, which are processed by concatenation, FC layer, and Softmax function in turn. The result of "ensemble prediction" can be formulated as

$$\begin{cases} \bar{y}_i = f_{\mathrm{ensemble}}\left(z_i^{(1)}, z_i^{(2)}, z_i^{(3)}\right) \\ \quad = \sigma\left(\mathrm{MLP}\left(z_i\right)\right) \\ z_i = \mathrm{Concat}\left(z_i^{(1)}, z_i^{(2)}, z_i^{(3)}\right) \end{cases} \tag{16}$$

where $\bar{y}_i \in \mathrm{R}^{N_c}$ denotes the ensemble predicted probability vectors over N_c categories. During the training of the classifiers ensemble, $\bar{y}_i$ is supervised by a one-hot GT label y_i through the standard cross-entropy (CE) loss, i.e.,

$$L_{\mathrm{CE}}(D) = \sum_{i=1}^{\mathrm{BS}} -y_i \log\left(\bar{y}_i\right). \tag{17}$$

In summary, the deep-supervised and self-supervised losses shown in (15), combined with the classifier's ensemble loss shown in (17), the "Joint Loss (JL)" used in the training of the MPFASS-Net model is defined as

$$L_{\mathrm{JL}}(D) = L_{\mathrm{CE}}(D) + L_{\mathrm{Total}}(D). \tag{18}$$

TABLE I
DETAILED INFORMATION OF THREE DATASETS

Dataset	Images per class	Class	Total	Size	Resolution
AID	220-400	30	10000	600x600	0.5-0.8m
NWPU	700	45	31500	256x256	0.2-30m
UCM	100	21	2100	256x256	0.3m

IV. EXPERIMENTAL RESULTS AND ANALYSIS

A. Datasets

To evaluate the performance of the designed MPFASS-Net model, we conducted experiments on three publicly available datasets, i.e., UCM dataset [42], AID dataset [43], and NWPU dataset [44], respectively. The detailed information corresponding to the three datasets is shown in Table I.

1) UCM Dataset: This dataset is a small dataset that extensively covers 21 types of scenes, such as agricultural, airplane, baseball diamond, beach, buildings, chaparral, dense residential, forest, freeway, golf course, harbor, intersection, medium density residential, mobile home park, overpass, parking lot, river, runway, sparse residential, storage tanks, and tennis courts. Each category of scene consists of 100 RGB images, for a total of 2100 images. In this dataset, each image has a size of 256×256 and a spatial resolution of 0.3 m.

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2) AID Dataset: This dataset is a large-scale aerial scene classification dataset that covers a wide range of 30 types of scenes, including airport, bare land, baseball field, beach, bridge, center, church, commercial, dense residential, desert, farmland, forest, industrial, meadow, medium residential, mountain, park, parking, playground, pond, port, railway station, resort, river, school, sparse residential, square, stadium, storage tanks, and viaduct. Each type of scene consists of 220–400 RGB images, with a total of 10 000 images. In this dataset, each image has a size of 600 × 600 and a spatial resolution of 0.5–0.8 m.

3) NWPU Dataset: This dataset is a publicly available aerial image scene classification dataset created by Northwestern Polytechnical University. It covers a wide range of 45 types of scenes, including airplane, airport, baseball diamond, basketball court, beach, bridge, chaparral, church, circular farmland, cloud, commercial area, dense residential, desert, forest, freeway, golf course, ground track field, harbor, industrial area, intersection, island, lake, meadow, medium residential, mobile home park, mountain, overpass, palace, parking lot, railway, railway station, rectangular farmland, river, roundabout, runway, sea ice, ship, snow Berg, sparse residential, stadium, storage tank, tennis court, terrace, thermal power station, and wetland. Each type of scene consists of 700 RGB images, with a total of 31 500 images. In this dataset, each image has a size of 256 × 256 and a spatial resolution of 0.2–30 m.

To fairly compare with other methods during the experiments, we follow the RSI scene classification communal experiment rule [45], where each dataset is divided in two different proportions to randomly construct the training and testing datasets, i.e., from the UCM dataset 50%/80% of the images were randomly selected for constructing the training set, while the remaining 50%/20% of the images were used for constructing the testing set. Similarly, 20%/50% of the images were randomly selected from the AID dataset to construct the training set, while the remaining 80%/50% of the images were used to construct the testing set, and 10%/20% of the images were randomly selected from the NWPU dataset to construct the training set, and the remaining 90%/80% of the images were used to construct the testing set.

B. Experimental Details and Evaluation Indicators

During all the experiments, the resolution of every image was scaled to 224 × 224, the Adam optimizer was used to optimize the models, the number of training epochs was set to 150, the initial learning rate was set to 3×10^{-5}, and a cosine decay strategy was used to adjust the learning rate during the model training. And the batch size is set to 64. In addition, ResNet50 was initialized with pretraining parameters trained by ImageNet-1K. In particular, all experiments in this article were performed on an image server, which contains 60 GB RAM, GPU RTX4090 24 GB, Linux + Python 3.8, Pytorch 2.0.0+ Cuda11.8. In order to obtain reliable experimental data, we repeated all experiments five times, and the final recorded experimental results consisted of two parts, i.e., mean and standard variance.

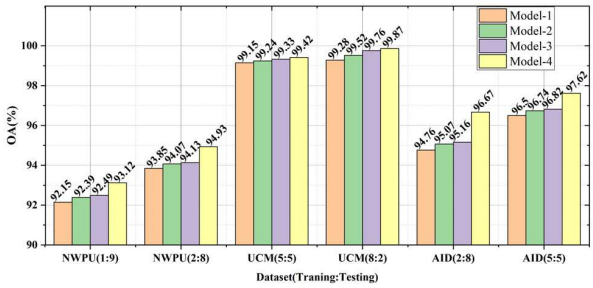


Fig. 5. Experimental results of ablation on the three datasets.

During the experiments, we used the overall accuracy (OA) and the confusion matrix (CM) as the evaluation metrics to measure the results of our experiments [12]. OA refers to the proportion of samples correctly classified by the model in all categories to the total sample, which can reflect the classification performance of the entire model [12]. CM presents the correspondence between the predicted results of a model on the test datasets and the true labels in the form of a table [45]. In addition, we also used Score-CAM [46] and T-SNE [47] to further validate the effectiveness of our designed model. Score-CAM utilizes the relationship between the final classification score and the feature map to generate a heatmap to display the model's focus in the classification decision area. T-SNE uses scatter plots to visually represent the distribution of features.

C. Ablation Experiments

1) MPFASS-Net Module Ablation Experiments: The designed MPFASS-Net model, as described in Fig. 2, consists of four modules, i.e., MSFE, PMSFA, DLCE, and OCSSL. To verify the effectiveness of each module, we designed four different submodels for module ablation experiments, i.e.,

1) *Model-1*: MSFE.
2) *Model-2*: MSFE + PMSFA.
3) *Model-3*: MSFE + PMSFA + OCSSL.
4) *Model-4*: MSFE + PMSFA + OCSSL + DLCE.

where *Model-1* is to utilizes only the 5th layer feature map (i.e., F_5) shown in (1), which is processed by GAP + FC + Softmax classifier to obtain the prediction results. Notably, *Model-1* is essentially a ResNet 50 network, which is chosen as the baseline in the experiments. *Model-2* further utilizes PMSFA for cross-layer feature fusion as shown in (2), then only based $AF_i^{(3)}$ to obtains the prediction result through the GAP + FC + Softmax classifier. *Model-3* is based on *Model-2* and introduces OCSSL to constrain the training of the model, i.e., further refine the fusion features using two self-supervised loss functions designed as shown in (12) and (13). *Model-4* means all modules are adopted, i.e., the MPFASS-Net designed in this article. Finally, the results of the ablation experiments on the three datasets are shown in Fig. 5.

Based on the ablation experiment results of classification accuracy shown in Fig. 5, we can directly draw the following conclusion, i.e., (1) *Model-4* has the highest classification accuracy, while *Model-1* has the lowest classification accuracy, and the classification accuracy of other models is between the two. (2) The designed PMSFA, OCSSL, and DLCE modules all play a positive role in improving the classification accuracy

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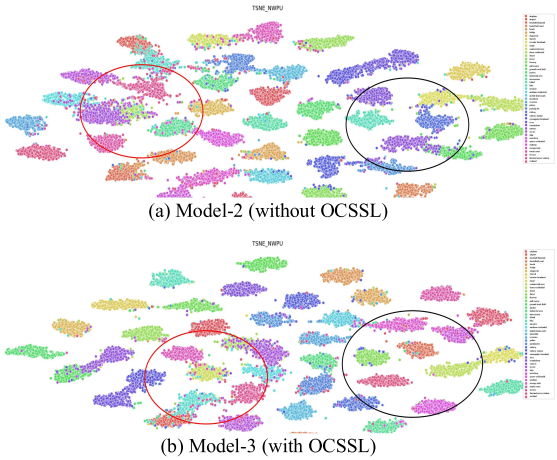
TABLE II
COMPARISON OF THE PARAMETERS OF FOUR SUBMODELS

Parameter	Model-1	Model-2	Model-3	Model-4
M	24.8435	31.9524	32. 0076	32.0626

of RSI scenes. Taking the UCM and AID datasets as examples, when the training and testing ratios are both set to 50%, the OA of *Model-1* is 99.15% and 96.5%, respectively. When the PMSFA module based on the DCAM-CLFA mechanism is introduced to *Model-1*, the OA of *Model-2* is improved by 0.09% and 0.24%, respectively. This indicates that the multiscale feature maps are interactively aggregated layer-by-layer through the DCAM-CLFA mechanism, which can better utilize the global contextual information of RSI to obtain multigranularity features that are helpful for RSI scene classification. Similarly, when the OCSSL module is introduced to *Model-2*, the OA of *Model-3* is improved by 0.09% and 0.08%, respectively, compared to *Model-2*. This indicates that the feature maps refined by OCSSL have better semantic discrimination ability and alleviate the semantic gap problem in RSI scene tasks. Finally, in *Model-4*, i.e., the MPFASS-Net designed in this article, the accuracy reached its maximum value, i.e., 99.42% and 97.62%. This indicates that the DLCE module is able to adaptively integrate classifiers from different levels of fused features, exploiting their complementarities and improving the model's classification ability.

Meanwhile, the parameter comparison results shown in Table II indicate that *Model*-2 has increased the parameter by 7.108 M compared to *Model-1*. This is because three DCAM-CLFA mechanisms were used in the PMSFA module, which increased the number of parameters. Compared to *Model-2*, *Model-3* has increased the number of parameters by 0.0552 M, which is due to the use of two SSL losses in OCSSL, resulting in an increase in the number of parameters from the class center matrix. Similarly, the number of parameters in *Model-4* increases by 0.055 M compared to that in *Model-3*, which is due to the need to use one FC layer of adaptive learning in the DLCE module for the integration of the three prediction results.

To visually demonstrate the effectiveness of OCSSL, we use the T-SNE [47], a nonlinear dimensionality reduction technique well-suited for projecting high-dimensional data into 2 or 3 dimensions, to visualize the multigranularity semantic features obtained by *Model-2* and *Model-3*. The visualization results are shown in Fig. 6(a) and (b). As shown in Fig. 6(a), each small cluster in the figure represents a category. Observing the figure, it can be seen that some small clusters in the figure have a larger cluster radius, indicating that the distribution within the clusters is relatively dispersed, i.e., the distribution of intraclass features is relatively dispersed. At the same time, the boundaries between some small clusters are relatively blurred, which means that the distribution between clusters is dense, and the overlap of features between classes is high. However, as shown in Fig. 6(b), the T-SNE figure represents the features processed by the OCSSL module. From this figure, it can be seen that the cluster radius of each cluster has decreased significantly compared to the cluster radius in



(a) Model-2 (without OCSSL)

(b) Model-3 (with OCSSL)

Fig. 6. T-SNE visualization results on the NWPU dataset. The red ellipse in (a) Represents the boundaries between clusters that are relatively blurred, indicating "small interclass variance." The black ellipse in (a) represents the distribution of features within the clusters is relatively scattered, i.e., "large intraclass variance." The red ellipse in (b) represents the significant reduction in boundary overlap between clusters. The black ellipse in (b) represents that the intracluster radius of each cluster is reduced.

Fig. 6(a). This means that after being processed by the OSSCL module, the distribution of intraclass features is more compact. At the same time, the overlap degree of some boundaries between clusters has been appropriately reduced compared to the overlap degree in Fig. 6(a). This means that after being processed by the OSSCL module, the distribution of interclass features is more dispersed. The above results fully demonstrate that our designed OCSSL module can alleviate the semantic gap problem of "large intraclass variance and small-inter class variance" in remote sensing scene images.

2) PMSFA Component Ablation Experiments: PMSFA, as the core module of the designed MPFASS-Net model, is mainly composed of the DCAM-CLFA mechanism, which aims to implement multiscale fusion at the feature level. As shown in Fig. 3, the DCAM-CLFA mechanism is mainly composed of three components, i.e., DA, MIE, and DCA cross-layer feature aggregation. Therefore, to further validate the effectiveness of these three components for feature fusion, based on the designed MPFASS-Net model, and simplifying the DCAM-CLFA mechanism of its PMSFA module, we constructed the following three networks for ablation experiments, i.e.,

1) *Net-1*: DA + DCA.
2) *Net-2*: DA + MIE.
3) *Net-3*: DA + MIE + DCA.

where *Net-1* indicates that only DA and DCA are used to implement cross-layer feature aggregation. *Net-2* indicates that only DA and MIE are used for feature aggregation, i.e., for S_1 and S_2 shown in (6), the feature aggregation is accomplished by adding them directly. While *Net-3* indicates that all three components are used for feature aggregation, i.e., the complete MDCA-CLFA mechanism shown in Fig. 3 is used. Finally, the ablation experiment results based on three datasets are shown in Fig. 7.

According to the ablation experiment results shown in Fig. 7, *Net-3* achieved the highest classification accuracy,

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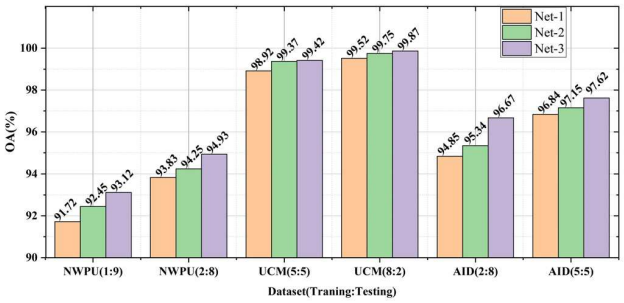


Fig. 7. Results of component ablation experiments based on three datasets for the PMSFA module.

followed by *Net-2*, while *Net-1* had the lowest classification accuracy. This is because Mamba has a stronger ability to model the global contextual information in RSI compared to DCA, and the designed DCAM-CLFA mechanism adds two mamba branch on the basis of the original DCA, it not only maintains the original global contextual information modeling ability of Mamba, but also couples it with the cross-layer interaction ability of DCA, which is more conducive to mining the complementarity between adjacent two-layer features and enhancing the discriminative ability of fused features.

D. Comprehensive Comparative Experiments

To comprehensively evaluate the performance of the designed MPFASS-Net model for RSI scene classification, we have selected various state-of-the-art models to conduct comprehensive comparison experiments, and the participating comparison methods can be roughly categorized into three types, i.e., 1) classical CNN-based methods, including ResNet50 [5], VGG16 [48], VSDNet [6], MBFANet [7], CGLINet [49], CASD [50], and SAGN [11]; 2) multilayer feature aggregation-based methods, including ACR-MLFF [13], CSCANet [14], CDLNET [16], SAF-Net [17], EMSCNet [18], PFFGCN [12], and MSAFEB [15]; and 3) transformer-based methods, including EMTCAL [51], HHTL [9], SCViT [10], ViT-B-16 [8], PSCLI-TF [2], LVM-StARS [19], and PVT-V2B [52]. The results of the experiments on the three datasets will be reported separately in the following.

1) Experiments on UCM Dataset: Table III shows the results of the comparison experiments between the designed MPFASS-Net model and the other three types of RSI classification models. From the comparison experimental results, it can be found that the designed MPFASS-Net achieves the highest classification accuracy, i.e., when the ratio of the training images is 50% and 80%, the designed model improves the classification accuracy compared to the best CNN-based method [i.e., CASD (50%), CGLINet (80%)] by 0.35% and 0.03%, respectively, and the classification accuracy is improved by 0.7% and 0.11% compared to the best feature aggregation-based method [i.e., CDLNET (50%), CSCANet (80%)], and the classification accuracy is improved by 0.52% and 0.25% compared to the best Transformer-based method [i.e., SCViT (50%), PSCLI-TF (80%)], respectively.

As shown in Table III, the designed MPFASS-Net model achieves the highest classification accuracy on UCM data. This is because RSI belongs to composite semantics, and to identify them, it must rely on local objects and their dependencies in the image. The designed PMSFA module based on the DCAM-CLFA mechanism can capture local key features from complex backgrounds in RSI, while also mining their global dependencies. After hierarchical cross-layer aggregation through PMSFA, it can capture the multigranularity semantic information of RSI. In addition, under the constraints of the designed OCSSL, the semantic gap problem of "large intraclass differences and small interclass differences" can be effectively alleviated, thereby greatly improving the classification accuracy of RSI.

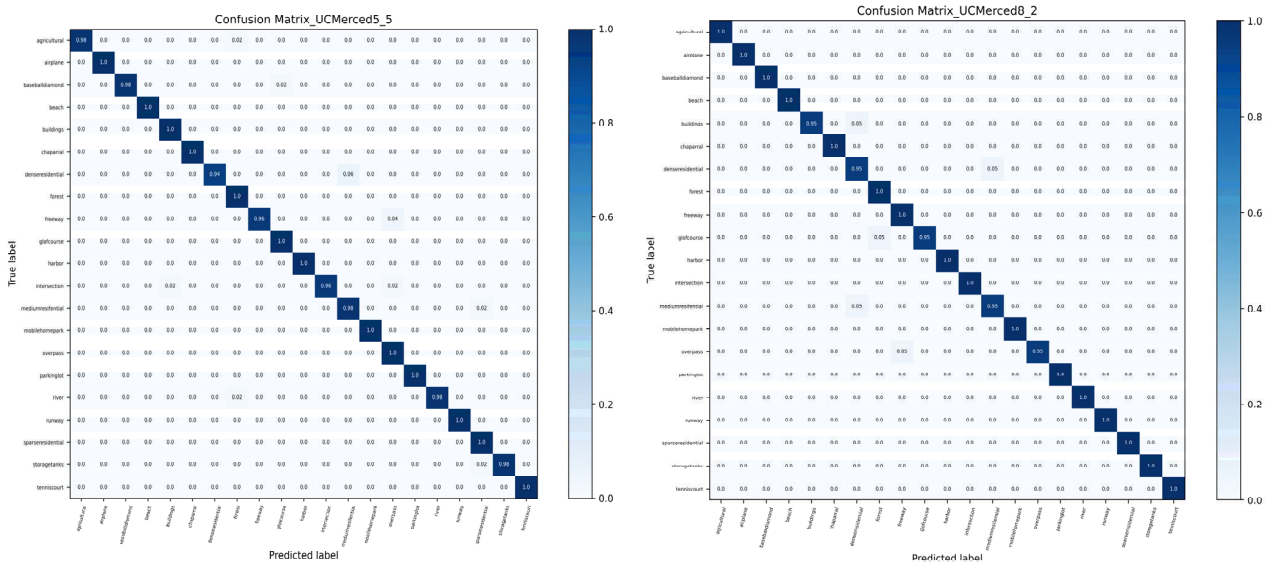
TABLE III
OA AND STANDARD DEVIATIONS (%) OF DIFFERENT METHODS ON THE UCM DATASET

Method Name	Years	Tr=50%	Tr=80%
ResNet50[5]	CVPR2016	95.33	98.33
VGG-16[48]	TGRS2015	94.14±0.69	95.21±1.20
VSDNet[6]	TGRS2022	98.49±0.18	99.67±0.12
MBFANet[7]	GRSL2023	-	99.66±0.19
CGLINet[49]	TGRS2024	-	99.84±0.16
CASD [50]	TGRS2024	99.07±0.09	99.62±0.11
SAGN[11]	TIP2023	-	99.76±0.12
ACR-MLFF[13]	GRSL2022	97.99±0.26	99.37±0.15
CSCANet[14]	GRSL2023	-	99.76±0.16
CDLNET[16]	TGRS2023	98.72±0.17	99.66±0.11
SAF-Net[17]	GRSL2024	-	99.32±0.16
EMSCNet[18]	TGRS2023	98.70±0.46	99.44±0.16
PFFGCN[12]	JSTARS2024	99.04±0.19	99.67±0.21
EMTCAL[51]	TGRS2022	98.67±0.16	99.57±0.28
HHTL[9]	JSTARS2022	98.87±0.28	99.48±0.28
SCViT[10]	TGRS2022	98.90±0.19	99.57±0.31
ViT-B-16[8]	LCLR2021	97.83±0.16	98.95±0.24
PSCLI-TF[2]	GRSL2024	98.70±0.16	99.62±0.21
PVT-V2B[52]	CVM2022	97.94±0.44	98.86±0.38
MPFASS-Net	Ours	**99.42±0.15**	**99.87±0.07**

In addition to OA, as shown in Fig. 8, we also use CM to demonstrate the specific classification accuracies of MPFASS-Net for each class of RSI under different training ratios. As shown in Fig. 8(a), when the training ratio is 50%, there are 13 categories with 100% classification accuracy, and 5 categories with 98% classification accuracy. However, since "dense residential" and "medium density residential" both belong to residential, there is a strong similarity between them, which can easily cause confusion. As shown in Fig. 8(b), when the training ratio is 80%, the classification accuracy of almost all scenes has reached 95% or above, and 16 out of the total 21 categories achieve 100% classification accuracy. These experimental results indicate that the designed MPFASS-Net model is a very effective classification method for the UCM dataset.

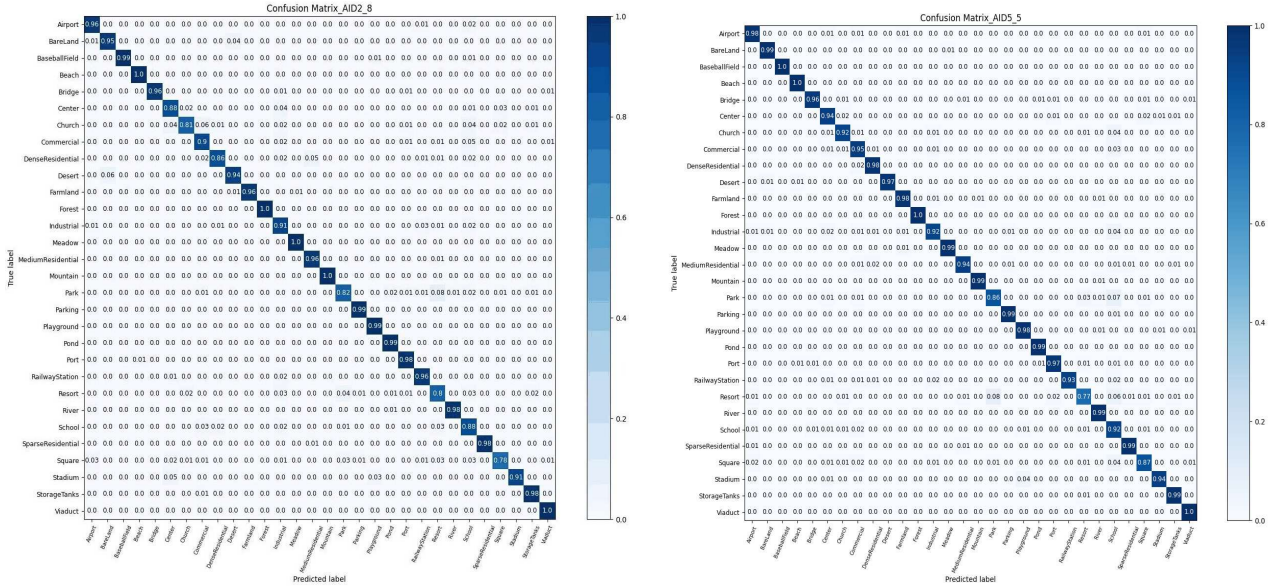
2) Experiments on AID Dataset: Table IV shows the classification results of different models on the AID dataset. Similar to the UCM dataset, the designed MPFASS-Net model achieves the best classification performance, i.e., when the ratio of the training images is 20% and 50%, the designed model achieves 96.67% and 97.62% classification accuracies, which is an improvement in OA compared to the second-best method by 0.39% and 0.1% (e.g., PSCLI-TF). The reason is that the designed MPFASS-Net model can effectively perceive the local key targets of RSI from complex backgrounds using

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(a) (b)

Fig. 8. Confusion matrices of the MPFASS-Net on the UCM dataset. (a) CM at 50% training image ratios. (b) CM at 80% training image ratios.



(a) (b)

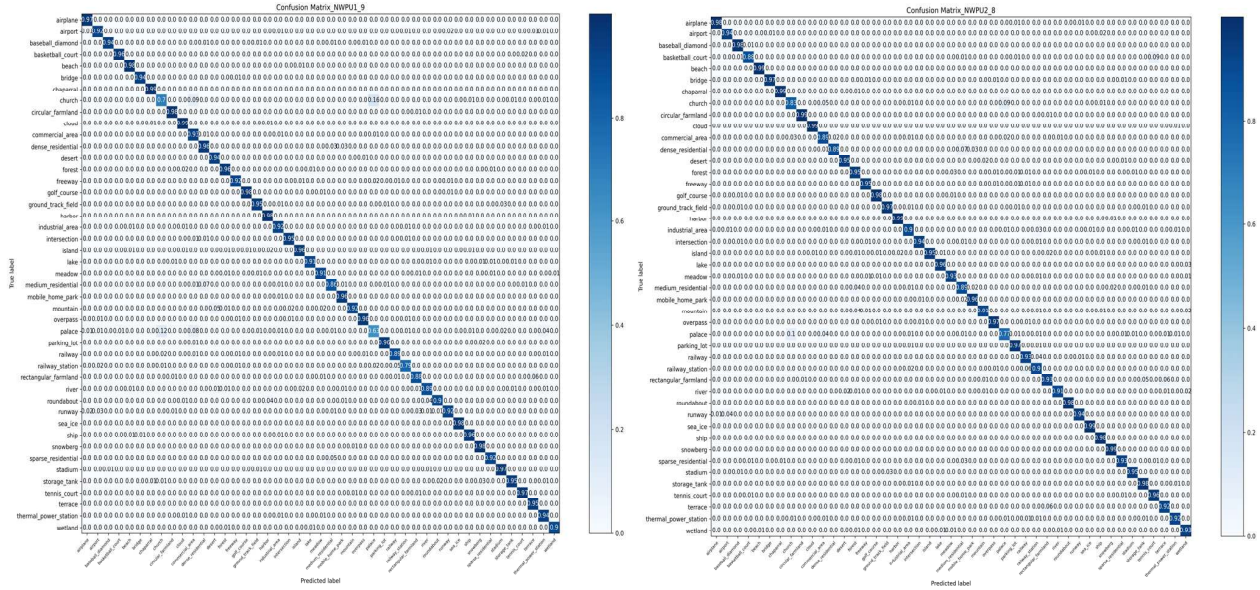
Fig. 9. Confusion matrices of the MPFASS-Net on the AID dataset. (a) CM at 20% training image ratios. (b) CM at 50% training image ratios.

MSFE, global context information mining, cross-layer feature aggregation, and SSL, so that the rich and diverse land cover in RSI scenes can be fully captured, thereby achieving the highest classification accuracy. Meanwhile, we also found an interesting phenomenon that by introducing attention (e.g., MBFANet), multiscale feature fusion (e.g., CDLNET), or global contextual information modeling (e.g., HHTL), the accuracy of RSI scene classification can be significantly improved compared to conventional CNN (e.g., ResNet50 and VGG-16) models. These phenomena also demonstrate that introducing multiscale feature aggregation and global contextual information modeling is feasible and effective in improving RSI classification accuracy.

In addition, the CM of our designed MPFASS-Net based on the AID dataset at different training ratios is shown in Fig. 9. When the training ratio is 20% and 50%, 23 and 27 out of the total of 30 classes have classification accuracies of more than 90%, respectively. Nevertheless, the classification accuracy of "square" is only 78% when the training ratio is 20%. This is because "square" contains open spaces and buildings, and there are similar key areas in scenarios such as "center, church, park, resort, etc.," which leads to a decrease in the model's recognition rate. However, when the training samples increase to 50%, the classification accuracy of this scenario improves to 87%.

3) Experiments on NWPU Dataset: Table V shows the classification results of different models on the NWPU dataset. It is evident that the designed MPFASS-Net model achieved the highest classification accuracy. When the training image ratio is 10% and 20%, the designed model has improved

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(a) (b)

Fig. 10. Confusion matrices of the MPFASS-Net on the NWPU dataset. (a) CM at 10% training image ratios. (b) CM at 20% training image ratios.

TABLE IV

OA AND STANDARD DEVIATIONS (%) OF DIFFERENT METHODS ON THE AID DATASET

Method Name	Years	Tr=20%	Tr=50%
ResNet50[5]	CVPR2016	88.17	92.94
VGG-16[48]	TGRS2018	86.59±0.29	89.64±0.36
VSDNet[6]	TGRS2022	96.00±0.18	97.28±0.14
MBFANet[7]	GRSL2023	93.98±0.15	96.93±0.16
CGLINet[49]	GRSL2024	95.35±0.14	97.10±0.24
CASD [50]	TGRS2024	95.72±0.13	96.96±0.11
SAGN[11]	TIP2023	95.35±0.19	96.90±0.23
ACR-MLFF[13]	GRSL2022	92.73±0.12	95.06±0.33
CSCANet[14]	GRSL2023	94.67±0.20	96.83±0.14
CDLNET[16]	TGRS2023	95.47±0.11	97.19±0.11
SAF-Net[17]	GRSL2024	94.20±0.12	96.72±0.14
EMSCNet[18]	TGRS2023	95.13±0.10	96.96±0.10
PFFGCN[12]	JSTARS2024	95.88±0.23	97.40±0.14
MSAFEB[15]	GRSL2023	92.24±0.02	95.85±0.03
EMTCAL[51]	TGRS2022	94.69±0.14	96.41±0.23
HHTL[9]	JSTARS2022	95.62±0.13	96.88±0.21
SCViT[10]	TGRS2022	95.56±0.17	96.98±0.16
ViT-B-16[8]	LCLR2021	93.74±0.27	95.84±0.29
PSCLI-TF[2]	GRSL2024	96.28 ±0.21	97. 52±0.15
LVM-StARS[19]	GRSL2024	95.70±0.07	97.62±0.24
PVT-V2B[52]	CVM2022	93.52±0.35	96.27±0.14
MPFASS-Net	Ours	**96.67±0.25**	**97.62±0.16**

TABLE V

OA AND STANDARD DEVIATIONS (%) OF DIFFERENT METHODS ON THE NWPU DATASET

Method Name	Years	Tr=10%	Tr=20%
ResNet50[5]	CVPR2016	85.94	88.47
VGG-16[48]	TGRS2018	76.47±0.18	79.79±0.15
VSDNet[6]	TGRS2022	92.13±0.16	94.68±0.13
MBFANet[7]	GRSL2023	91.61±0.14	94.01±0.08
CGLINet[49]	TGRS2024	92.28±0.17	94.38±0.14
CASD [50]	TGRS2024	92.28±0.12	94.75±0.13
SAGN[11]	TIP2023	91.75±0.22	93.53±0.21
ACR-MLFF[13]	GRSL2022	90.01±0.33	92.45±0.20
CSCANet[14]	GRSL2023	91.27±0.11	93.72±0.10
CDLNET[16]	TGRS2023	92.28±0.25	94.50±0.19
SAF-Net[17]	GRSL2024	90.94±0.08	93.62±0.01
EMSCNet[18]	TGRS2023	92.16±0.07	94.08±0.20
PFFGCN[12]	JSTARS2024	92.91±0.15	94.89±0.12
MSAFEB[15]	GRSL2023	91.70±0.002	94.09±0.002
EMTCAL[51]	TGRS2022	91.63±0.19	93.65±0.12
HHTL[9]	JSTARS2022	92.07±0.44	94.21±0.09
SCViT[10]	TGRS2022	92.72±0.04	94.66±0.10
ViT-B-16[8]	LCLR2021	90.05±0.29	92.61±0.14
PSCLI-TF[2]	GRSL2024	92.92 ±0.20	94.86±0.16
LVM-StARS[19]	GRSL2024	92.48±0.02	94.14±0.10
PVT-V2B[52]	CVM2022	89.72±0.16	92.95±0.009
MPFASS-Net	Ours	**93.12±0.12**	**94.93±0.11**

OA values by 0.84% or 0.18% compared to the best CNN-based method (i.e., CASD), 0.84% and 0.43% compared to the best feature aggregation-based method (i.e., CDLNET), and 0.20% and 0.07% compared to the best Transformer-based method (i.e., PSCLI-TF), respectively. Due to the fact that the NWPU dataset contains more images and categories, although it is more challenging compared to the UCM and AID datasets, the designed MPFASS-Net can still achieve encouraging classification accuracy, once again proving the superiority of the ideas and methods we adopted in our model design.

In addition, the CM corresponding to designed MPFASS-Net under different training ratios based on the NWPU dataset are shown in Fig. 10. Observing the results of two CMs, it can be found that when the training ratio is 10% and 20%, 38 and 39 out of the total 45 categories have classification accuracies above 90%, which achieves excellent classification performance over most of the categories. However, the “church” and “palace” categories in the NWPU dataset are easily confused

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TABLE VI
OA AND STANDARD DEVIATIONS (%) OF DIFFERENT BACKBONES ON THE THREE DATASETS

Backbone	UCM		AID		NWPU	
	50%	80%	20%	50%	10%	20%
ResNet18	99.28±0.34	99.56±0.17	96.32±0.25	97.14±0.21	92.87±0.32	94.76±0.27
ResNet34	99.36±0.27	99.67±0.11	96.53±0.18	97.35±0.19	92.99±0.25	94.88±0.22
ResNet50	99.42±0.15	99.87±0.07	96.67±0.15	97.62±0.16	93.12±0.12	94.93±0.11
VGG16	98.92±0.38	99.56±0.25	96.07±0.33	96.99±0.23	91.50±0.43	94.56±0.40
ViT-B-16[8]	99.39±0.12	99.71±0.10	96.60±0.16	97.55±0.17	93.01±0.15	94.91±0.12

with each other, this is because both of them have similar architectural structures and scene layouts, with extremely high similarities in texture structure and color features.

E. Backbone Network Robustness Experiments

Table VI shows the integration performance of our designed MPFASS-Net with different backbone networks (i.e., ResNet18, ResNet34, ResNet50, VGG16, and ViT-B-16 [8]) on the three datasets. When choosing ResNet as the backbone network for the model, as the number of layers in the backbone network increases from 18 to 50, the classification results of the MPFASS-Net model show subtle changes, but the overall classification results are relatively stable. This indicates that the number of layers in the backbone network has a relatively small impact on the classification performance of the model, demonstrating the robustness of the MPFASS-Net model in the integration of backbone networks. While the backbone network switches from ResNet to VGG and ViT, the classification accuracies of the MPFASS-Net model on all three datasets are improved compared to their original accuracies (i.e., the accuracies shown in Tables III–V). This indicates that the designed PMSFA, OCSSL, and DLCE modules have strong robustness to the backbone network, and can improve their performance in RSI scene classification tasks.

F. Computational Complexity Evaluation

To more intuitively reflect the comparison of computational complexity between the designed MPFASS-Net model and other SOTA models, we conducted benchmark testing on the scatter plot depicting OA. The floating-point operations per second (FLOPS) and the parameters (Params) are widely used metrics for evaluating computational costs. In this article, the results of FLOPS and Params from MPFASS-Net are derived from the dynamic training process of the model on a 224×224 sized image on the AID dataset. OA reveals the classification performance of the model on the AID test set, while the remaining SOTA results are sourced from relevant literature. The experiment results are shown in Fig. 11, which depicts the scatter plot of OA as a function of Params and FLOPS. It reveals that compared to the designed model, LVM-STARS achieves comparable OA but requires nearly 2.5 times the computational complexity. Although CASD, CGLINet, HHTL, PFFGCN, and SCViT have lower computational complexity, they cannot achieve optimal classification performance. CDLINet has the highest computational complexity, but its OA is basically the same as PFFGCN's OA and still has a certain gap compared to MPFASS-Net. Although PVT-V2B and ResNet50 have relatively low computational complexity, their classification performance lags significantly behind other methods. The above discussion fully confirms that the proposed MPFASS-Net outperforms all methods in OA and does not bring additional computational burden to the inference process of the model.

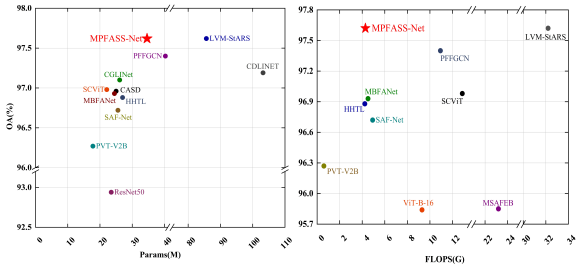


Fig. 11. OA versus Params and FLOPs on the AID dataset.

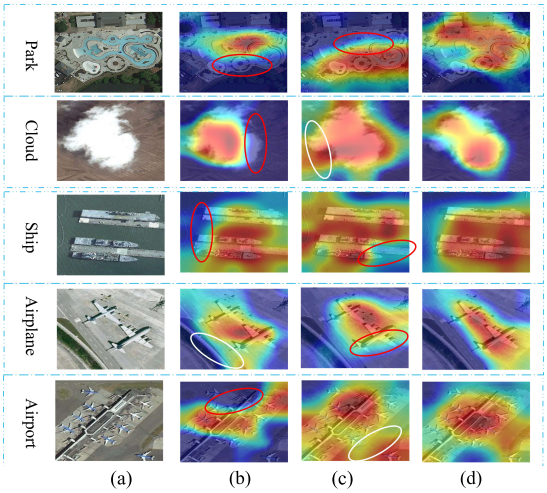


Fig. 12. Heat maps of the visualization results for different methods on the NWPU dataset with 20% training ratios. The red ellipse represents missed feature extraction. The white ellipse represents the extraction of incorrect features. (a) Original image. (b) ResNet50 [50]. (c) ViT-B-16 [8]. (d) MPFASS-Net.

G. Visualization Experiments

To visually observe and analyze whether the designed MPFASS-Net model really focuses on the key targets or regions of interest in the image when performing RSI scene classification, as shown in Fig. 12, we use the Score-CAM [17] method to visualize the features of the ResNet50 [5], ViT [8], and MPFASS-Net models, i.e., the importance of each neuron in the last feature map of the model is computed using gradient and class activation map (CAM) methods to locate the key regions that influence the model decisions, using images from

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the NWPU dataset. In the heat map, redder regions indicate that they are more discriminative and are regions of more interest to the model.

From the visualization results shown in Fig. 12, it can be seen that the designed MPFASS-Net model can more accurately and completely focus on key targets and regions of interest in RSI compared to baseline methods such as ResNet50 and ViT. For example, the first row of the image belongs to the "park" scene, and its semantics are mainly expressed by a single central area of the image, although other methods can also locate the region of interest, while the designed MPFASS-Net model focuses on a more accurate and complete region of interest. The third row of images belongs to the "ship" scene, and its semantics are clearly reflected in two key areas in the image. Obviously, the designed MPFASS-Net model pays more attention to these two key areas, while the ViT method loses one area of interest. Meanwhile, the image in the fifth row belongs to the "airport" scene. Although its key targets are not concentrated in a single area of the image and are scattered in various positions of the image, from the heatmap corresponding to MPFASS-Net, it focuses on a wider and more complete area, while other methods are affected by background interference and fall into local attention. This is because in the designed MPFASS-Net model, the multiscale local features from ResNet50, after the DCAM-CLFA mechanism, have both local perception and global contextual information mining ability, which enables it to accurately locate the key targets from complex backgrounds, and thus has stronger discriminative ability.

V. Conclusion

In order to further improve the accuracy of RSI scene classification, a novel MPFASS-Net model is designed to address the semantic gap problem caused by the complex background of RSI and the large-scale variation of the target of interest.

The designed model mainly consists of four modules, i.e., MSFE, PMSFA, OCSSL, and DLCE. The ablation experiment results based on three standard datasets show that:

1) The DCAM-CLFA mechanism constructed by embedding the DCA attention into mamba, the ablation experiments shown in Fig. 5 indicate that it has better feature extraction capability than the original mamba encoder.

2) After PMSFA based on the DCAM-CLFA mechanism for multiscale features, the ablation experiments shown in Fig. 7 show that there will be a better ability to represent multigranularity features of RSI.

3) The designed OCSSL function can effectively alleviate the semantic gap problem in RSI scene classification.

In summary, the designed MPFASS-Net model outperforms other state-of-the-art methods in comparative experiments based on three standard RSI scene classification datasets due to its stronger multigranularity feature perception and generalization capabilities. However, it also has the following limitations, i.e., MPFASS-Net is only a scene classification method based on single modal RSI, and cannot further improve classification accuracy by utilizing the complementarity of multimodal data such as HSI and LiDAR. In addition, the DCAM-CLFA mechanism and deep-supervised classifiers ensemble, which increases the number of parameters in the model, and its feature fusion and model generalization capabilities have not been evaluated in other applications. Therefore, in future work, we will consider extending MPFASS-Net to a multimodal remote sensing data classification method, and also will further explore new Mamba encoding methods, design new feature fusion and classifiers ensemble schemes to further improve the performance of the model in RSI scene classification, and extend the research results to other image recognition tasks to verify the generalization ability of the designed model.

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